Lightweight Multiscale Neural Architecture Search With Spectral–Spatial Attention for Hyperspectral Image Classification

Chunhong Cao, Han Xiang, Wei Song, Hongbo Yi, *Member, IEEE*, Fen Xiao, *Member, IEEE*, and Xieping Gao, *Member, IEEE*

***Abstract*— Hyperspectral image (HSI) classification based on neural architecture search (NAS) is a currently attractive frontier as it not only automatically searches complex neural network architecture but also avoids professional knowledge and experience design and alleviates the lacking of generalization ability as well when dealing with a new classification task. However, the existing HSI classification based on NAS has some drawbacks: 1) a huge number of training parameters and high calculations are inductive to overfitting and high complexity and 2) efficient operators are lacking in the search space, which can distinguish spatial locations and spectral features in different bands. Furthermore, as the category samples in HSI data show a serious long-tail distribution phenomenon, HSI classification remains challenging. To address these issues, we propose a lightweight multiscale NAS with spatial–spectral attention (LMSS-NAS) for HSI classification. The main work includes threefold: 1) in order to reduce the number of model parameters and promote spectral–spatial feature fusion, a new lightweight efficient search space is designed, which consists of three equivalent lightweight convolution operators with multiple receptive fields; 2) to fully use the spectral–spatial correlation of HSI, a cube-to-pixel classification framework is designed to mine the local spatial and spectral context; and 3) focal loss and label smoothing loss in computer vision tasks are jointly migrated to LMSS-NAS to improve the unbalanced samples' classification and model robustness. Experimental results on four public hyperspectral datasets show that the proposed method can achieve competitive classification performance as well as low computational cost. The code is available at https://github.com/xh-captain/LMSS-NAS.**

Manuscript received 23 December 2022; revised 12 February 2023 and 22 February 2023; accepted 28 February 2023. Date of publication 6 March 2023; date of current version 16 March 2023. This work was supported in part by the Natural Science Foundation of Hunan Province of China under Grant 2022JJ30552 and Grant 2022JJ30571, in part by the Research Foundation of Education Department of Hunan Province of China under Grant 21A0109 and Grant 21B0172, in part by the National Natural Science Foundation of China (CN) under Grant 61972333 and Grant 62272404, and in part by the National Key Research and Development Program of China under Grant 2020YFA071350. *(Corresponding authors: Han Xiang; Xieping Gao.)*

Chunhong Cao, Han Xiang, Wei Song, Hongbo Yi, and Fen Xiao are with the MOE Key Laboratory of Intelligent Computing and Information Processing, Xiangtan University, Xiangtan 411105, China (e-mail: caoch@xtu.edu.cn; 202021002370@smail.xtu.edu.cn; 202021002361@smail.xtu.edu.cn; 201705551212@smail.xtu.edu.cn; xiaof@xtu.edu.cn).

Xieping Gao is with the Hunan Provincial Key Laboratory of Intelligent Computing and Language Information Processing, Hunan Normal University, Changsha, Hunan 410081, China (e-mail: xpgao@xtu.edu.cn).

This article has supplementary downloadable material available at https://doi.org/10.1109/TGRS.2023.3253247, provided by the authors.

Digital Object Identifier 10.1109/TGRS.2023.3253247

***Index Terms*— Hyperspectral image (HSI) classification, lightweight, neural architecture search (NAS), spatial separable convolution (SSC), spectral–spatial attention.**

I. Introduction

HYPERSPECTRAL images (HSIs) provide more discriminative features for ground object observation than other common remote images as they collect rich spatial and spectral information reflecting various features of ground objects. Therefore, HSIs have been widely applied in geological exploration, target detection, and landscape classification [1], [2] where identifying the category of each pixel in HSI has become the critical technique [3], [4]. However, HSI classification suffers from insufficient spatial resolution and limited labeled samples, which may lead to unstable and low classification accuracy.

Recently, many deep learning (DL)-based methods have been applied in HSI classification and achieved remarkable progress. In the early years, most feature extraction methods in DL were implemented by handcrafted feature extractors, and two main categories are included, i.e., spectral information-based and spectral–spatial information-based classification methods. The former focused only on spectral features ignoring spatial correlation resulting in poor performance [5], such as stacked autoencoder (SAE) [6], deep belief network (DBN) [7], and convolutional neural network (CNN), 1-D CNN [8]. As known, different objects in HSI may show similar spectral features and the same objects in different locations exhibit different spectral features such that approaches using the spectral information alone in classification tasks will produce unsmooth classification map as many discrete pixels may be misclassified. The latter one usually obtains better classification accuracy as more structure information is considered in the methods compared with the former. Accordingly, many spectral–spatial classification methods were proposed, such as spectral–spatial residual network (SSRN) [9], 3-D CNN [10], [11], Dense-Net [12], and full convolutional neural network (FCN) [13]. Besides, Hamida et al. [14] introduced a new 3-D DL approach that enables a joint spectral and spatial information process.

However, in the spectral–spatial methods, one can observe that the classification accuracy of HSI is sensitive to the

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:27:38 UTC from IEEE Xplore. Restrictions apply.

network scale design as the scales of ground objects in homogeneous regions are diverse in real scenario [15]. Therefore, to better characterize the variability of spatial features at different scales, multiscale spatial features and 1-D spectral features were jointly learned from HSI in M3D-DCNN [16]. Lee and Kwon [17] used a novel deep CNN architecture that could jointly optimize the spectral and spatial information of HSI. Hao et al. [18] exploited a Gaussian curvature filter-based feature extraction method with multiscale segmentation constraint. A hybrid multiscale network combining 3-D CNN with 2-D CNN was proposed by Roy et al. [19], which significantly improved the classification performance, while these multiscale models also brought more training parameters resulting in more computation complexity.

Considering the HSI's real-time processing demand for intelligent analysis of remote sensing, an important task is to explore the lightweight classification model that can balance the high classification accuracy and low computational complexity for HSI classification. Inspired by the successful application of the lightweight design of GoogLeNet [20], spatial separable convolution (SSC) [21], and GhostNet [22] in computer vision, lightweight models were attracted increasing attention for HSI classification with limited samples. Jia et al. [23] presented a lightweight CNN (LWCNN) for the small sample classification of HSIs, which adopted a spectral–spatial Schroedinger eigenmap (SSSE) feature extraction to obtain joint spatial–spectral information and significantly reduced the number of parameters. A deep 3-D lightweight convolution network (3D-LWNet) [24] was proposed in 2019 and applied to the HSI classification ensuring fewer training parameters and computational complexity. Xue et al. [25] exploited a lightweight spectral–spatial network (SSN) to efficiently extract the spectral–spatial features of HSI with small parameter size and designed a novel spectral–spatial Siamese network (S3Net) for few-shot HSI classification. Meng et al. [26] designed a lightweight spectral–spatial convolution module (LS^{2}CM) as an alternative to the convolutional layer, which greatly reduced network parameters and computational complexity while maintaining or even improving the classification performance. Lv et al. [27] proposed an efficient spectral–spatial interaction network (ESSINet), which designed an involution-based residual block to incorporate spatial and spectral information. Cui et al. [28] established a 3-D depthwise convolution-based architecture, LiteDepthwiseNet (LDWNet), which decomposed standard convolution into depthwise convolution and pointwise convolution to reduce the model complexity with well classification performance. It is noted that although the lightweight network effectively reduces the computing requirements, it may compromise with some performance in classification.

To alleviate the above issue, attention mechanism [29], [30], [31] was introduced to HSI's feature extraction. For example, residual spectral–spatial attention networks (RSSANs) [32] employed a spectral–spatial attention layer in SSRN to select and enhance spectral and spatial information. Xue et al. [33] proposed a novel attention-based second-order pooling network (A-SPN) to model the spectral–spatial information and discriminative features. Roy et al. [34] proposed an attention-based adaptive spectral–spatial kernel improved residual network (A^2S^2K-RSNet) with spectral attention to capturing discriminative spectral–spatial features for HSI classification. A novel local transformer with spatial partition restore network (SPRLT-Net) was proposed by Xue et al. [35], which can capture global context dependencies and adapt the inherent changes of HSI spatial pixels. Combining GhostNet and attention mechanism, Wang et al. [36] proposed a neural architecture search (NAS)-guided lightweight spectral–spatial fusion neural network (LMAFNet), which was guided by several conclusions of NAS [37], and exploited multiscale ghost grouped with efficient channel attention (ECA) [38] module for adaptively adjusting the weights of different channels. However, the abovementioned models were handcrafted, which relied on the experts' professional knowledge and experience heavily. For high-dimensional HSI data with hundreds of spectral bands and limited labeled samples, the deeper network structure, the more difficult it is to design the optimal network structure. Furthermore, a variety of hyperparameters need to be tuned, such as convolution kernel size, branches, or jump connections. This undoubtedly brings serious challenges to design the general classification models for the diverse HSI data.

Fortunately, increasing interests were rising on developing solutions to automate the manual architectural design. For example, NAS [39], [40], [41], [42] has achieved promising performance in many tasks such as image classification and object detection. Specifically, Chen et al. [44] introduced differentiable architecture search (DARTS) [43] into HSI classification for the first time and proposed Auto-CNN. Subsequently, Zhang et al. [45] proposed a 3-D asymmetric NAS (3D-ANAS) model and designed a 3-D asymmetric convolutional search space based on DARTS, which greatly improved the classification performance. However, such 3-D convolution increased the model parameters and calculation significantly such that the network with powerful representation capacity may suffer from overfitting. Accordingly, studying NAS-based method for HSI classification remains challenging.

Obviously, researchers generally utilized attention mechanism with different scales to extract effective features for diverse ground objects avoiding small target's misclassification in the HSI classification. However, there is no similar mechanism in automatically search model for HSI classification yet. Furthermore, various samples in HSI dataset show a long-tail distribution, resulting in the unbalanced results in HSI classification.

To address the above issues, in this article, we propose a lightweight multiscale NAS with spatial–spectral attention for HSI classification, named LMSS-NAS. Specifically, we first design new lightweight and efficient operators for the search space from two main concerns: 1) in order to improve the classification accuracy and reduce computational complexity, a multiscale equivalent SSC operation is designed, where the large-scale filters are replaced by small-scale filters while keeping the same receptive field and 2) considering the small target's misclassification, the lightweight spectral–spatial

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:27:38 UTC from IEEE Xplore. Restrictions apply.

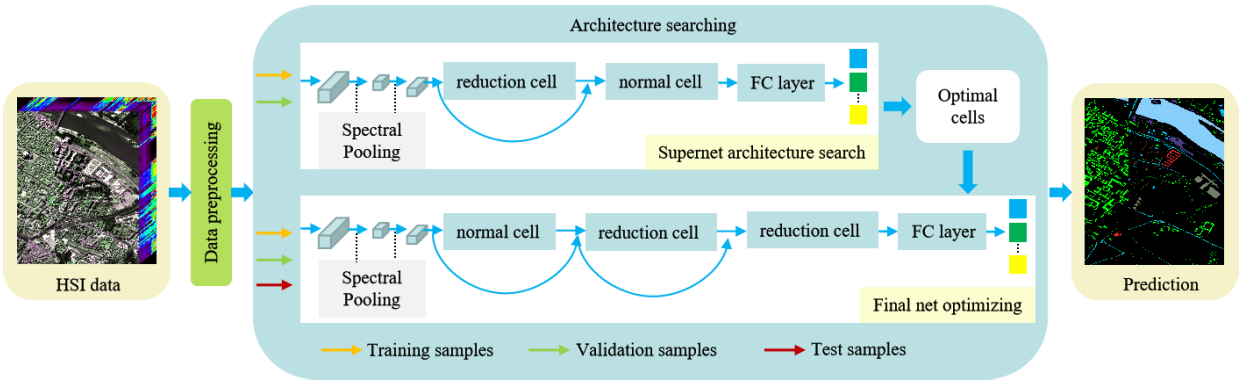


Fig. 1. Framework of the proposed LMSS-NAS for HSI classification.

attention method from convolutional block attention module (CBAM) [31] is introduced to strengthen the significant spectral–spatial information and suppress redundant information as well as the noise. Then, we build a DARTS-based HSI classification network under the cube-to-pixel framework by considering both spectral–spatial correlation and the long-tail distribution in HSI. The main contributions can be summarized as follows.

1) By analyzing the characteristics of HSI, a new lightweight efficient search space is designed, which consists of three equivalent lightweight convolution operators with spectral–spatial attention mechanism.
2) The proposed LMSS-NAS under the cube-to-pixel classification framework greatly reduces the model's parameters and promotes the ability of spectral–spatial feature fusion for fast HSI classification.
3) By integrating weighted label smoothing loss into the focal loss (FL), the proposed model achieves balanced accuracy as the FL and the label smoothing loss can deal with unbalanced samples caused by the long-tail distribution and the model generalization as well.

Experiments are conducted on four typical HSI datasets, including Houston University (HU), Kennedy Space Center (KSC), Pavia University, and Pavia Center, showing more competitive classification results with fewer training parameters and computations. At the same time, the classification results by crossing datasets confirm the generalization ability of the proposed LMSS-NAS. The remainder of this article is organized as follows. Section II describes our approach in detail. Section III reports and discusses the experimental results on four benchmark hyperspectral datasets. Finally, this work is summarized in Section IV.

II. Proposed Method

In this article, a lightweight automatic search network LMSS-NAS is proposed for HSI classification. The overall architecture is shown in Fig. 1, which consists of two stages roughly: data preprocessing and NAS. First, considering the low spatial resolution of HSI, cube-to-pixel label information pairs were established to fully obtain the relevant spectral–spatial information of HSI. Second, NAS includes supernet architecture search and final net optimizing, where we design lightweight search operators for search space and build an efficient cell for supernet architecture. At the same time, in order to balance the classification accuracy of samples with the long-tail distribution in HSI, the weighted loss function was proposed by combining FL and label smoothing loss to accelerate model convergence and alleviate the unbalanced results. The detail is described in the following.

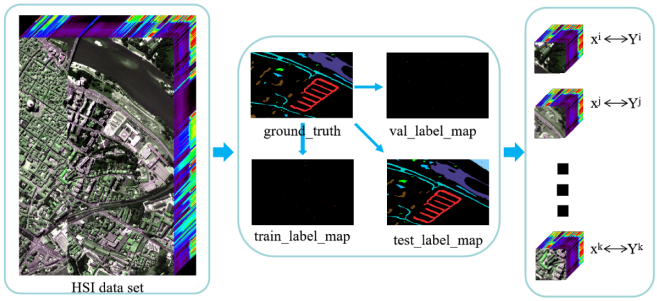


Fig. 2. Example of 3-D Cube-to-Pixel sample extraction in the Pavia Center dataset.

A. Data Preprocessing

HSIs have rich spectral information with a strong correlation between adjacent bands but low spatial resolution. Accordingly, in order to make full use of the correlation between spatial and spectral dimensionalities, we design the cube-to-pixel classification framework. First, the training sample set is divided according to each pixel's label. To illustrate the procedure explicitly, the example is shown in Fig. 2. In each category, the samples of given number are taken as training samples and validation samples from the HSI, and the remaining samples are acted as test samples. To ensure the data's natural distribution, the samples are selected in a random way. In order to strengthen the spectral–spatial correlation of training samples, according to each selected labeled pixel, the corresponding neighborhood cube $\in \mathbb{R}^{S\times S\times B}$ was extracted from the original hyperspectral data as the actual input to form a training pair (cube, pixel), i.e., (X^i, Y^i) as shown in Fig. 2, where S and B are the size of the spatial window and spectral bands of the dataset, respectively.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:27:38 UTC from IEEE Xplore. Restrictions apply.

B. Neural Architecture Search

1) Search Space: As HSI classification based on DARTS first needs to determine the search space, we present the search space $O = \{o^{(i,j)}\}$, which contains the following operations.

1) None: $f(x) = 0$.
2) avg_pool_3 × 3: AvgPool2d.
3) max _pool_3 × 3: MaxPool2d.
4) skip_connect: $f(x) = x$.
5) M_{CS}_sepConv_(3 × 3): $M_{\text{CS}} - \text{DR} - \text{SSC}(3 \times 3) - \text{BN} - M_{\text{CS}}$.
6) M_{CS}_sepConv_(5 × 5): $M_{\text{CS}} - \text{DR} - \text{SSC}(3 \times 3) - \text{BR} - \text{SSC}(3 \times 3) - \text{BN} - M_{\text{CS}}$.
7) M_{CS}_sepConv_(7 × 7): $M_{\text{CS}} - \text{DR} - \text{SSC}(3 \times 3) - \text{BR} - \text{SSC}(3 \times 3) - \text{BR} - \text{SSC}(3 \times 3) - \text{BN} - M_{\text{CS}}$.

Here, DR is Dropout(0.5) and Relu, BR is batch normalization (BN) and Relu, SSC is spatial separable convolution, and M_{CS}_sepConv_$(k \times k)$ $(k = 3, 5, 7)$ are the proposed operators, which help to extract deeper spectral–spatial characteristics, and the rest operators are commonly used in NAS-based methods. The designing motivations are explained from the following two aspects.

a) Multiscale equivalent SSC: Convolution is a commonly used operation in NAS-based HSI classification. Previous work included conventional convolution (CC) [43], [44], depthwise SC [43], [44], [45], and dilated convolution (DC) [43], [44]. It can be seen that various convolutions led to different computations and parameters. High-dimensional convolution will bring large parameters and time consumption, which makes it difficult to build deeper models. Therefore, the high-dimensional convolution is generally converted into low-dimensional SC. For example, suppose that x with the size of $C_{\text{in}} \times H \times W$ is acted as the input and $C_{\text{out}} \times H \times W$ feature needs to be obtained, where C_{in} and C_{out} denote the input and output spectral bands, respectively, and the number of parameters P_1 and the floating-point operations (FLOPs) F_1 in $k \times k$ CC (i.e., $\text{Conv}(k \times k) = \text{conv}(k \times k)(x)$) are computed by

$$P_1 = C_{\text{in}} \times k \times k \times C_{\text{out}} \tag{1}$$

$$F_1 = C_{\text{in}} \times k \times k \times C_{\text{out}} \times H \times W. \tag{2}$$

Auto-CNN [44] used SC and DC in the search space, where SC is expressed as $\text{SepConv}(k \times k) = \text{conv}(1 \times 1)(\text{conv}(k \times k)(x))$, and the parameters and FLOPs are computed as

$$P_2 = C_{\text{in}} \times k \times k + 1 \times 1 \times C_{\text{in}} \times C_{\text{out}} \tag{3}$$

$$F_2 = C_{\text{in}} \times k \times k \times H \times W + 1 \times 1 \times C_{\text{in}} \times C_{\text{out}} \times H \times W. \tag{4}$$

DC is expressed as $\text{DilConv}(k \times k) = \text{conv}(1 \times 1)(\text{conv}(k \times k)(x))$, and d voids are added to $k \times k$ convolution to obtain a receptive field of $(k + d) \times (k + d)$. Even though the receptive field is expanded, the parameters and FLOPs remain P_2 and F_2, respectively. Compared with CC, SC and DC add pointwise convolution after depthwise convolution to extract spatial and spectral features in turn. Taken $C_{\text{in}} = C_{\text{out}} = C$, the parameters and FLOPs are just $(1/k^2 + 1/C)$ of CC. In [45], 3D-ANAS adopted 3-D convolution with depthwise separable, where convolution mode is expressed as $\text{SepConv3d}(k \times k \times k) = \text{conv}(k \times 1 \times 1)(\text{conv}(1 \times k \times k)(x))$. The parameters and FLOPs are computed as

$$P_3 = C_{\text{in}} \times 1 \times k \times k + k \times 1 \times 1 \times C_{\text{in}} \times C_{\text{out}} \tag{5}$$

$$F_3 = C_{in} \times k \times k \times 1 \times H \times W + k \times 1 \times 1 \times C_{\text{in}} \times C_{\text{out}} \times H \times W. \tag{6}$$

Similarly, the parameters and FLOPs are $(1/k + 1/C)$ of CC.

Obviously, Auto-CNN obtained the automatic classification based on DARTS, but its accuracy needs to be improved. The 3D-ANAS obtained better classification results through a novel search space design, but the model convergence was very slow due to the greatly increased parameters and FLOPs. To address this dilemma between the high accuracy and complexity, we introduce the SSC from Inception-v3 [21], which is expressed as $\text{Spatial_sepConv}(k \times k) = \text{conv}(1 \times 1)(\text{conv}(1 \times k)(\text{conv}(k \times 1)(x)))$. It requires only P_4 parameters and F_4 FLOPs, which are $(2k + C)/(C + k^2)$ of the SC, where

$$P_4 = 2 \times (C_{\text{in}} \times k \times 1) + 1 \times 1 \times C_{\text{in}} \times C_{\text{out}} \tag{7}$$

$$F_4 = 2 \times (C_{\text{in}} \times k \times 1 \times H \times W) + 1 \times 1 \times C_{\text{in}} \times C_{\text{out}} \times H \times W. \tag{8}$$

Besides, due to the diversity of ground objects' scale in HSI scenes, the extraction of multiscale spectral–spatial features is one of the keys to improve the performance of HSI classification. In previous works, different scale convolutions are usually used in the search space. Multiscale convolutions of 3 × 3, 5 × 5, and 7 × 7 are designed in Auto-CNN [44] and 3D-ANAS [45]. However, as analyzed in the above context, the parameters and computation are positively correlated with the size of input data and convolution kernel. Thus, we replace large-scale filters with small filters to reduce the parameters while maintaining large-scale receptive fields.

As shown in Fig. 3, 5 × 5 and 7 × 7 convolution can be equivalently substituted by superposing two and three conventional 3 × 3 convolution operations, respectively. Furthermore, 3 × 3 CC can be equivalently replaced with 3 × 3 SSC. It can be seen that the parameters and computation are further reduced. The detailed comparison of parameters and FLOPs for different convolutions is shown in Table I. When $C_{\text{in}} = C_{\text{out}} = C$, taken 3 × 3 CC superposition to replace the original 5 × 5 and 7 × 7 convolution, the parameters and computation of the two convolutions are reduced by 27% and 45%, respectively. When using 3 × 3 SSC for equivalent substitution, the parameters are reduced by 52% and 63%, and FLOPs are $12/(25C)$ and $18/(49C)$.

b) Attention-based search space: To enhance the adaptive feature extraction ability of the designed convolution operations, the lightweight attention module CBAM [31] is adopted to strengthen the cube's spectral and spatial adaptive learning ability. Spectral channel attention M_C and spatial attention M_S are expressed as follows:

$$\begin{aligned} M_C(F) &= \sigma(\text{MLP}(\text{AvgPool}(F) + \text{MLP}(\text{MaxPool}(F)) \\ &= \sigma\left(W_1\left(W_0\left(F^C_{\text{avg}}\right)\right) + W_1\left(W_0\left(F^C_{\text{max}}\right)\right)\right) \\ F' &= M_C(F) \otimes F \end{aligned} \tag{9}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:27:38 UTC from IEEE Xplore. Restrictions apply.

TABLE I
COMPARISON OF PARAMETERS AND FLOPS FOR DIFFERENT CONVOLUTION REPLACEMENT

Operation	Convolution	Repeat	Parameters	FLOPs
$conv(3\times3)$	$conv(3\times3)$	–	$9\times C^2$	$9\times C^2\times H\times W$
$conv(5\times5)$	$conv(5\times5)$	–	$25\times C^2$	$25\times C^2\times H\times W$
$conv(7\times7)$	$conv(7\times7)$	–	$49\times C^2$	$49\times C^2\times H\times W$
$conv(5\times5)$	$conv(3\times3)$	2	$18\times C^2$	$18\times C^2\times H\times W$
$conv(7\times7)$	$conv(3\times3)$	3	$27\times C^2$	$27\times C^2\times H\times W$
$Spatial_sepconv_(3\times3)$	$conv(1\times3)(conv(3\times1))$	–	$6\times C^2$	$6\times C\times H\times W$
$Spatial_sepconv_(5\times5)$	$conv(1\times3)(conv(3\times1))$	2	$12\times C^2$	$12\times C\times H\times W$
$Spatial_sepconv_(7\times7)$	$conv(1\times3)(conv(3\times1))$	3	$18\times C^2$	$18\times C\times H\times W$

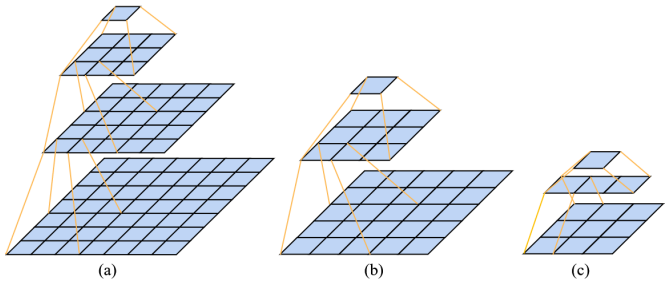


Fig. 3. Mini-network replacing. (a) Replacing 7×7 convolution with three layers of 3×3 convolution. (b) Replacing 5×5 convolution with two layers of 3×3 convolution. (c) Replacing 3×3 convolution by a 3×1 convolution followed by a 1×3 convolution.

$$\begin{aligned} M_S(F) &= \sigma\left(f^{7\times7}\left(\left[\text{AvgPool}\left(F'\right);\text{MaxPool}\left(F'\right)\right]\right)\right) \\ &= \sigma\left(f^{7\times7}\left[F'^{S}_{\text{avg}};F'^{S}_{\text{max}}\right]\right) \\ F'' &= M_S\left(F'\right)\otimes F' \end{aligned} \tag{10}$$

where $\otimes$ denotes element-wise multiplication, σ denotes the sigmoid function, $W_0\in\mathbb{R}^{C_{\text{in}}/r\times C_{\text{in}}}$, $W_1\in\mathbb{R}^{C_{\text{in}}\times C_{\text{in}}/r}$, and $f^{7\times7}$ represents 7×7 convolution. The spectral–spatial CBAM is defined as $M_{\text{CS}}(F)=M_S(M_C(F))$.

2) Model Regularization and Search Strategy: They are given as follows.

a) Model regularization: Due to the multiple categories involved in HSI classification, the sample quantity in each category exhibits a long-tail distribution, leading to the extremely low classification accuracy of some categories in the classification tasks. Therefore, this article proposes a fusion loss of combined FL and label smoothing loss (SM) to balance the classification accuracy of various categories.

As shown in formula (11), in essence, FL [46] is based on cross entropy (CE) by introducing $(1-p_t)^\gamma$ to control the weight of different categories, where $\gamma\geq0$, y denotes the label of x with K categories, and y_i is the probability of each category of label y

$$\begin{aligned} L_{\text{focal}} &= -(1-p_t)^\gamma\sum_{i=1}^{K}y_i\log q_i \\ y_i &= \begin{cases}1, & \text{if } (i=y)\\ 0, & \text{if } (i\neq y),\end{cases} \qquad p_t=\exp\left(\sum_{i=1}^{K}y_i\log q_i\right) \\ q_i &= \frac{\exp(x_i)}{\sum_{j=1}^{K}\exp\left(x_j\right)}. \end{aligned} \tag{11}$$

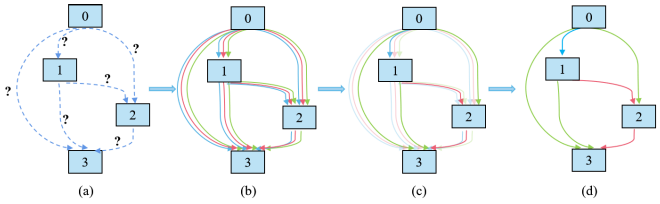


Fig. 4. DARTS search process. (a) Operations between nodes are initially unknown. (b) Placing mixed candidate operations between nodes. (c) Operational probability optimization update for hybrid operation. (d) Inducing the final architecture from the learned mixing probabilities.

However, it can be seen from (11) that when a sample is misclassified, p_t takes the minimum value and $(1-p_t)^\gamma$ is close to 1, while when p_t value is close to 1, $(1-p_t)^\gamma$ is close to 0. At this time, the model will assign a full probability to each training sample resulting in overfitting and lack of generalization ability. Inspired by the idea of [21], overfitting and model generalization ability can be improved by building SM. Therefore, this article constructs a fusion loss function (12) by integrating FL with SM

$$\begin{aligned} L_{\text{fs}} &= L_{\text{focal}}+L_{\text{smooth}} \\ L_{\text{smooth}} &= -\sum_{i=1}^{K}y_i\log q_i, \quad y_i=\begin{cases}(1-\varepsilon), & \text{if } (i=y)\\ \dfrac{\varepsilon}{K-1}, & \text{if } (i\neq y)\end{cases} \end{aligned} \tag{12}$$

where ε is hyperparameters, set to 0.05 in this article.

b) Search strategy: In this article, DARTS [43] is used to design the network architecture, as shown in Fig. 1, where the reduction cell and the normal cell are included. Multiple intermediate nodes are set in the cell as the intermediate feature extractor, and its structure is shown in Fig. 4. Let n_i be the node in the cell, and C_l of the cell is output by connecting the intermediate node, which is expressed as follows:

$$\overline{o}^{(i,j)}\left(n_j\right)=\sum_{o\in O}\frac{\exp\left(\alpha_o^{(i,j)}\right)}{\sum_{o'\in O}\exp\left(\alpha_{o'}^{(i,j)}\right)}o\left(n_j\right) \tag{13}$$

$$n_i=\sum_{j<i}\overline{o}^{(i,j)}\left(n_j\right) \tag{14}$$

$$C_l=\sum_{i}\text{concat}(n_i) \tag{15}$$

where $\alpha_o^{(i,j)}$ is the connection weight of $o^{(i,j)}$ between n_i and n_j. After optimizing the supernet, the final cell is built

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:27:38 UTC from IEEE Xplore. Restrictions apply.

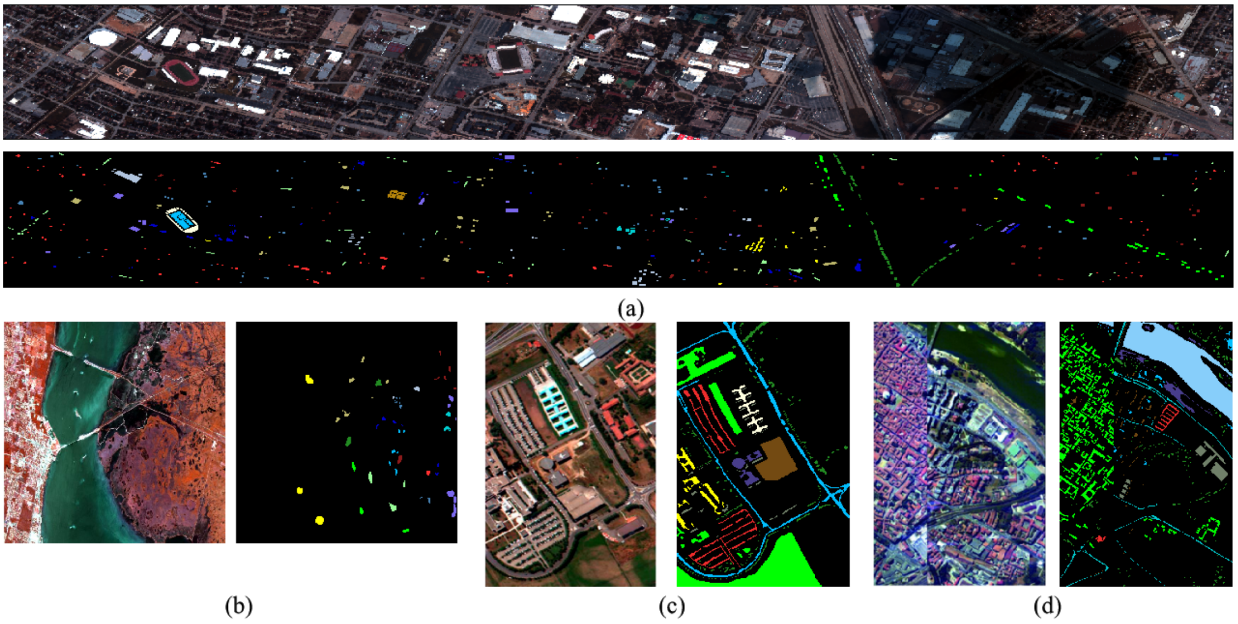


Fig. 5. False-color composites and ground-truth maps of HSI datasets. (a) HU. (b) KSC. (c) PU. (d) PC.

TABLE II
ATTRIBUTE ANALYSIS OF FOUR HYPERSPECTRAL DATASETS

Data Set	Houston University		KSC		Pavia University		Pavia Center	
No.	Category	Labelled samples	Category	Labelled samples	Category	Labelled samples	Category	Labelled samples
1	Healthy Grass	1251	Scrub	761	Asphalt	6631	Water	65971
2	Stressed Grass	1254	Willow swamp	243	Meadows	18649	Trees	7598
3	Syntheic Grass	697	CP hammock	256	Gravel	2099	Asphalt	3090
4	Trees	1244	Slash pine	252	Trees	3064	Self-Blocking Bricks	2685
5	Soil	1242	Oak/Broadleaf	161	Painted metal sheets	1345	Bitumen	6584
6	Water	325	Hardwood	229	Bare Soil	5029	Tiles	9248
7	Residential	1268	Swamp	105	Bitumen	1330	Shadows	7287
8	Commmercial	1244	Graminoid marsh	431	Self-Blocking Bricks	3682	Meadows	42826
9	Road	1252	Saprtina marsh	520	Shadows	947	Bare Soil	2863
10	Highway	1227	Cattail marsh	404				
11	Railway	1235	Salt marsh	419				
12	Parking Lot 1	1233	Mud flats	503				
13	Parking Lot 2	469	Water	927				
14	Tennis Court	428						
15	Running Track	660						
	Total	15029	Total	5211	Total	42776	Total	148152

according to $\overline{o}^{(i,j)} = \arg\max_{o\in O}\alpha_o^{(i,j)}$, $\alpha = \{\alpha_o^{(i,j)}|i, j, o \in O\}$ and the supernet weight ω in cells is optimized alternately according to the training set and validation set, as shown in formula (16) and (17), where $L_{\text{fs}}^{\text{train}}$ and $L_{\text{fs}}^{\text{val}}$ are the fusion loss functions calculated by training and validation samples according to (12), respectively

$$\alpha^* = \min_\alpha L_{\text{fs}}^{\text{val}}\left(\omega^*, \alpha\right) \tag{16}$$

$$\text{s.t. } \omega^*(\alpha) = \arg\min_\omega L_{\text{fs}}^{\text{train}}(\omega, \alpha). \tag{17}$$

III. Experiment and Analysis

In this section, we verify the validity of the proposed method on four real HSI datasets. First, we describe the datasets and implementation details in Sections III-A and III-B, respectively. Second, we compare the performance of LMSS-NAS with state-of-the-art approaches using qualitative metrics, including overall accuracy (OA), average accuracy (AA), and kappa coefficient (Kappa) in Section III-C. Third, in order to demonstrate the complexity of LMSS-NAS, we present the comparison results in terms of the involved training parameters, FLOPs, GPU memory usage, test time, and inference speed. Finally, to investigate the effectiveness of the designed search space and regularization, we evaluate the performance of the proposed method through ablation study.

A. Datasets

The HU dataset was acquired by the ITRES CASI-1500 sensor and presented by IEEE GRSS Data Fusion Contest 2013. The KSC dataset was acquired by NASA's AVIRIS sensor at the KSC in Florida in 1996. The University of Pavia (PU) and the Centre of Pavia (PC) datasets were gathered by the Reflective Optical System Imaging Spectrometer (ROSIS) sensor during a flight over Pavia, northern Italy, in 2001. The details of the datasets[1] are shown in Tables II and III. False-color images and ground-truth maps of the datasets are shown in Fig. 5.

[1]https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:27:38 UTC from IEEE Xplore. Restrictions apply.

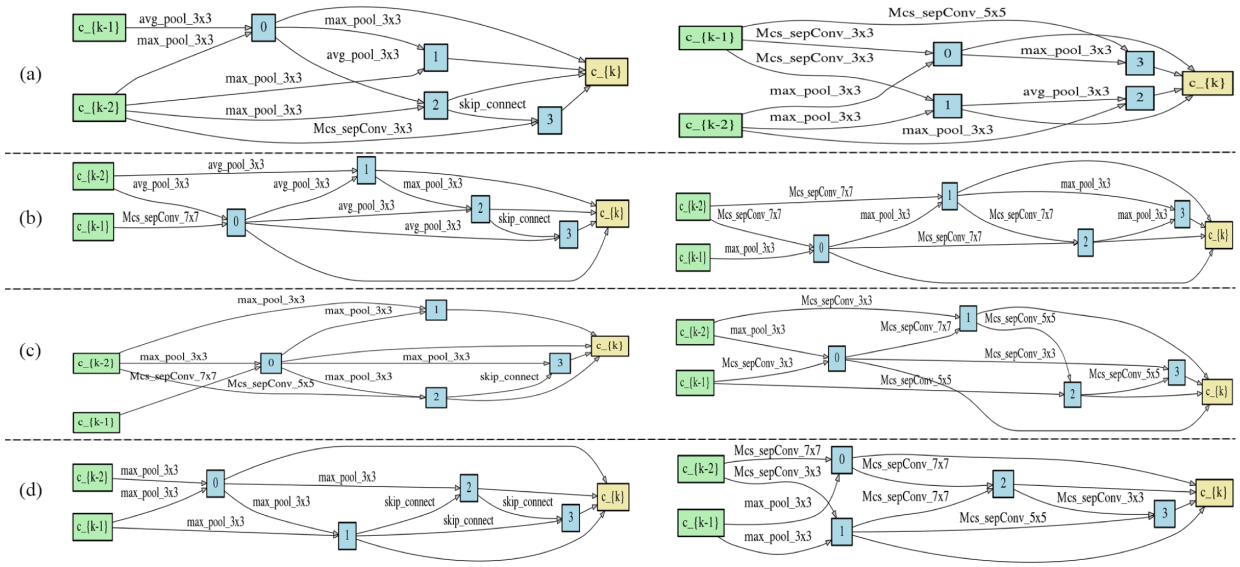


Fig. 6. Optimal structure of cells based on the LMSS-NAS method on four datasets, the left is normal cell, and the right is reduction cell. (a) HU. (b) KSC. (c) PU. (d) PC.

B. Implementation Details

For dataset sampling, we follow the way in [45] to select the labeled samples and conduct experiments in two different settings in view of fair comparison, i.e., 1) 20 and ten labeled samples are randomly selected from each category to build the training set and validation set, and the rest is used as the test set, and 2) 30 labeled samples are used for training samples for each class, keeping the validation and test set same as 1). In order to ensure the fairness and stability of comparison, each experiment is repeated ten times and the average is taken as the final result. The proportion of the number of training samples to the original data is shown in Table IV.

To promote the ability of spectral–spatial feature fusion for fast HSI classification, we construct the supernets for four datasets where S and B in the input cube $\in \mathbb{R}^{S\times S\times B}$ are set to 27 and the corresponding spectral bands, respectively. We have compared the proposed algorithm with different window sizes (more detailed results can be seen in Tables VII–X as well as the visualized results in Figs. 1–12 in the Supplementary Material). Both the training batch size and infer batch size are set to 32, the Adam optimizer is used to optimize the architecture parameters with the default weight decay = 0.0003, the learning rate = 0.004, the grad clip = 5, the drop prob = 0.2, and the momentum = 0.9. For supernet architecture search and final network optimization, we iteratively optimize the network for 100 and 500 iterations, respectively. In the model sampling setting of the comparison method, all methods use overlap sampling mode to calculate the pixels in the specified window of each labeled sample except for 1D-AT-CNN and 3D-ANAS. Specifically, the window sizes of 3D-LWNet, 3D-AT-CNN, LS^2CM, and S3Net are set to 27 × 27, 32 × 32, 11 × 11, {13 × 13, and 7 × 7}, respectively, while EESINet, LDWNet, LMAFNet, and A^2S^2K-RSNet are set to 9 × 9. The hyperparameters of the comparison methods (e.g., window size) are set according to the corresponding articles. Considering the different setting in terms of the best window sizes in the comparison methods, we also give the results under the same window size in the comparison methods (the details can be seen in Tables I–VI in the Supplementary Material).

TABLE III

ATTRIBUTE ANALYSIS OF FOUR HSI DATASETS

Data set	Size	labeled samples	Category	Resolution(m)
HU	349 × 1905 × 144	15029	15	2.5
KSC	512 × 614 × 176	5211	13	18
PU	610 × 340 × 103	42776	9	1.3
PC	1096 × 715 × 102	148152	9	1.3

TABLE IV

PARTITION SETTINGS FOR DATASET IN EXPERIMENT

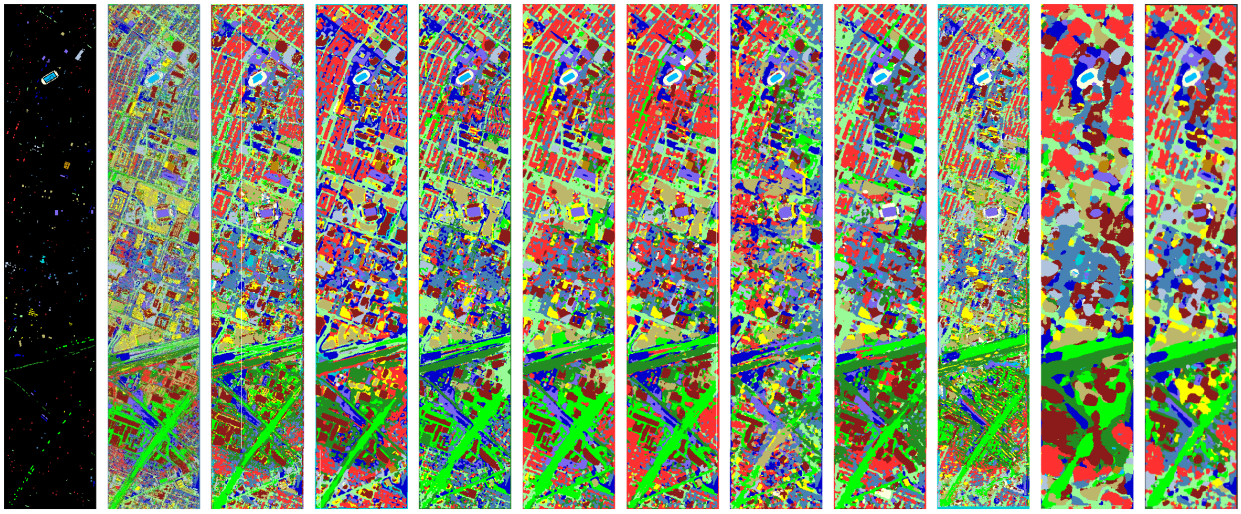
Setting	Data set	Train	Val	Test	Proportion(%)
20 labeled samples/class	HU	300	150	14579	1.99
	PU	180	90	42506	0.42
	PC	180	90	147882	0.12
	KSC	260	130	4821	4.9
30 labeled samples/class	HU	450	150	14429	2.99
	PU	270	90	42416	0.63
	PC	270	90	147792	0.18
	KSC	390	130	4691	7.4

As shown in Fig. 6, the cells with optimal architecture on each dataset are searched by the proposed LMSS-NAS, which includes normal cell and reduction cell. It can be seen that the architecture connection and operation in the cells searched for each dataset are diverse, where the normal cell is more inclined to choose the pooling operation without changing the size of the feature map, while the reduction cell prefers to choose the convolution operation to extract deep-level features and remove redundant information. The architectural differences searched by the respective datasets also reflect the adaptive performance of LMSS-NAS to target data.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:27:38 UTC from IEEE Xplore. Restrictions apply.

TABLE V
EXPERIMENTAL RESULTS ON THE HU DATASET

Models	1D-AT -CNN [44]	3D-ANAS [45]	LMAF -Net [36]	A^2S^2K -RSNet [34]	EESI -Net [27]	LDW -Net [28]	LS^2CM [26]	S3Net [25]	3D-LW -Net [24]	3D-AT -CNN [44]	LMSS -NAS	1D-AT -CNN	3D-ANAS	LMAF -Net	A^2S^2K -RSNet	EESI -Net	LDW -Net	LS^2CM	S3Net	3D-LW -Net	3D-AT -CNN	LMSS -NAS
Set.	20 labeled samples/class											30 labeled samples/class										
1	69.98	82.06	94.59	89.31	92.98	89.86	85	81.02	79.16	84.01	**97.42**	72.25	96.16	94.36	96.58	97.02	95.16	91.08	94.91	84.81	87.5	**97.73**
2	60.19	79.17	97.6	**98.58**	87.4	98.36	84.12	92.12	71.61	85.65	89.86	63.96	84.31	**99.04**	98.63	85.22	98.28	89.34	92.92	80.22	77.91	96.29
3	77.3	96.1	99.87	99.39	97.86	**99.73**	96.93	99.3	94.4	93.86	96.48	76.61	86.06	99.88	**99.98**	97.7	98.01	98.48	99.67	93.45	92.74	99.30
4	49.02	81.71	94.41	**98.75**	93.77	98.6	77.85	98.24	74.25	66.77	92.1	51.29	79.74	94.63	95.29	91.00	**96.50**	83.95	88.08	79.74	72.65	91.03
5	83.09	94.39	**99.20**	94.91	94.97	94.97	86.61	99.34	90.37	93.83	87.59	82.25	97.11	99.9	95.3	90.64	95.15	91.52	99.99	90.9	96.14	97.32
6	50.46	99.32	**99.25**	92.14	93.53	94.49	88.44	94.44	84.92	80.43	82.34	55.32	87.46	**98.74**	92.7	94.60	95.56	95.72	97.4	81.44	84.86	81.92
7	30.93	81.58	**95.27**	93.77	85.82	90.86	37.92	79.48	84.73	72.21	83.48	34.20	77.79	**93.67**	92.39	82.04	92.30	51.07	92.63	87.83	73.03	92.30
8	50.53	70.29	75.88	91.36	66.31	96.77	36.16	70.02	52.01	70.96	**99.11**	62.11	78.75	82.49	94.59	69.93	**99.72**	47.97	78.50	63.69	76.64	98.57
9	46.58	76.27	81.37	81.1	74.6	82.32	26.01	77.91	70.47	71.18	**84.95**	49.84	73.57	84.91	84.36	80.47	86.72	43.08	86.14	77.5	71.10	**88.77**
10	69.73	96.49	87.39	74.33	84.79	73.47	50.99	98.27	92.15	96.43	**98.40**	64.48	95.15	94.19	84.06	75.83	84.62	77.57	**98.18**	93.26	96.67	91.99
11	50.06	73.36	94.49	93.43	76.71	96.82	70.53	93.8	89.39	93.73	**98.72**	50.23	91.45	95.85	94.93	82.33	96.12	82.39	**98.15**	81.04	92.31	97.19
12	63.44	75.39	70.88	94.93	72.25	**96.20**	46.33	80.61	60.04	87.95	91.98	62.47	87.61	88.8	95.83	72.82	**98.70**	60.77	87.95	79.4	91.48	97.60
13	53.35	90.66	92.05	85.29	93.26	80.08	88.34	94.26	86.99	84.9	**96.48**	50.92	91.12	93.66	88.6	94.43	80.04	89.98	**96.62**	90.41	85.8	91.78
14	76.73	**100.00**	99.47	96.11	99.90	98.03	99.27	100.00	92.99	92.99	98.35	76.07	99.75	99.87	95.24	100.00	98.87	99.61	**100.00**	90.65	90.65	97.60
15	54	86.51	99.89	98.86	99.78	97.42	93.02	**99.98**	92.83	88.3	83.70	62.94	93.65	99.94	98.77	99.76	98.02	98.61	**100.00**	92.47	88.85	94.62
OA	58.35	83.15	90.67	90.94	85.37	91.31	65.73	88.96	78.95	83.37	**91.84**	60.36	86.16	93.75	93.3	85.21	93.87	75.95	92.93	84.17	84.45	**94.61**
AA	59.03	85.55	92.11	92.15	87.59	**92.39**	71.17	90.59	81.09	84.21	92.06	61	87.31	94.66	93.82	87.59	94.25	80.08	94.08	85.12	85.22	**94.27**
Kappa	55.18	81.81	89.92	90.21	84.18	90.78	62.98	87.98	77.32	82.07	**91.18**	57.35	85.03	93.24	92.75	84.01	93.37	74.01	92.36	82.96	83.25	**94.17**



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k) (l)

Fig. 7. Comparison of experimental results on HU using 30 training samples in each category. (a) Ground-truth map. (b) 1D-AT-CNN, window size (WS) = 1. (c) 3D-ANAS, WS = 1. (d) LMAFNet, WS = 9. (e) A^2S^2K-RSNet, WS = 9. (f) EESINet, WS = 9. (g) LDWNet, WS = 9. (h) LS^2CM, WS = 11. (i) S3Net, WS = {13,7}. (j) 3D-LWNet, WS = 27. (k) 3D-AT-CNN, WS = 32. (l) LMSS-NAS, WS = 27.

C. Classification Results

To evaluate the effectiveness of LMSS-NAS, we compare with 3D-LWNet[2] [24], A^2S^2K-RSNet[3] [34], LMAFNet[4] [36], LS^2CM[5] [26], S3Net[6] [25], EESINet [27], LDWNet [28] (handcrafted methods), 1D-AT-CNN, 3D-AT-CNN[7] [44], and 3D-ANAS[8] [45] (NAS-based methods) on four representative HSI datasets where the results of the other methods are obtained through the public reproducible codes.

The average OA, AA, and Kappa on HU, KSC, PU, and PC are listed in Tables V–VIII and the visualized results are shown in Figs. 7–10, and we give the labeled classification maps in Figs. 14–16 in the Supplementary Material. As can be seen, on HU, KSC, and PU datasets, the proposed LMSS-NAS achieves the best performance with respect to OA, AA, and Kappa as well as the balanced class accuracy as the minimum accuracy is higher than other methods. For the PC dataset, LMSS-NAS is almost as high as 3D-ANAS. The corresponding full classification maps of four datasets are visually shown in Figs. 7–10, where LMSS-NAS yields more accurate and smooth results. At the same time, as shown in Tables I–X as well as Figs. 1–13 in the Supplementary Material, LMSS-NAS yields clearer classification results and higher classification accuracy under the same window size than the competitors.

When 20 training samples are extracted per class, as seen in Table V, on HU, our method achieves 91.84% in terms of OA, and the seven handcrafted models, 3D-LWNet, LMAFNet, A^2S^2K-RSNet, LS^2CM, S3Net, EESINet, and LDWNet achieve 78.95%, 90.67%, 90.94%, 65.73%, 88.69%, 85.37%, and 91.31%, respectively. LMSS-NAS improves by 33.49%, 7.94%, and 8.16% in terms of OA compared with NAS-based 1D-AT-CNN, 3D-AT-CNN, and 3D-ANAS, respectively. As shown in Table VI, on KSC, our method achieves the highest OA 99.70%. LMAFNet (99.44%) and A^2S^2K-RSNet (99.24%), which combined attention and convolution, perform much better than 3D-LWNet (94.04%), 1D-AT-CNN (89.43%), 3D-AT-CNN (98.89%), LS^2CM (91.14%), S3Net (99.42%), and EESINet (97.02%). As shown in Table VII, on PU, LMSS-NAS achieves a superior OA of 96.78%. By contrast, the best OA of the NAS-based models is 96.53% obtained by 3D-ANAS. Also, compared with other methods

[2]https://github.com/hkzhang91/LWNet
[3]https://github.com/suvojit-0 × 55aa/A2S2K-ResNet
[4]https://github.com/githubltqc/LMAFN
[5]https://github.com/zhe-meng/LSSCM
[6]https://github.com/ZhaohuiXue/S3Net
[7]https://github.com/YushiChen/Auto-CNN-HSI-Classification
[8]https://github.com/hkzhang91/3D-ANAS

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:27:38 UTC from IEEE Xplore. Restrictions apply.

TABLE VI

EXPERIMENTAL RESULTS ON THE KSC DATASET

Models	1D-AT -CNN	3D-ANAS	LMAF -Net	A^2S^2K -RSNet	EESI -Net	LDW -Net	LS^{2}CM	S3Net	3D-LW -Net	3D-AT -CNN	LMSS -NAS	1D-AT -CNN	3D-ANAS	LMAF -Net	A^2S^2K -RSNet	EESI -Net	LDW -Net	LS^{2}CM	S3Net	3D-LW -Net	3D-AT -CNN	LMSS -NAS
Set.	20 labeled samples/class											30 labeled samples/class										
1	90.34	94.32	99.44	100.00	97.40	99.96	96.21	99.89	84.25	99.66	**100.00**	86.12	97.39	99.24	100.00	94.70	100.00	97.74	99.75	91.22	99.78	**100.00**
2	85.73	99.06	97.75	98.75	96.06	99.95	95.35	**100.00**	95.39	100.00	99.95	85.81	99.80	99.75	100.00	98.52	**100.00**	99.11	100	97.62	100.00	99.95
3	94.69	87.38	99.2	99.48	99.60	**100.00**	96.73	97.83	94.38	100.00	99.96	89.07	93.33	99.26	99.91	97.13	**100.00**	95.28	100	98.26	100.00	99.72
4	59.50	75.63	95.00	99.38	90.18	97.49	74.01	96.98	89.41	100.00	**100.00**	76.93	90.52	96.23	**99.81**	93.68	98.89	87.12	99.67	86.45	96.08	99.71
5	69.39	95.72	99.62	100.00	92.67	99.08	90.46	**100.00**	100.00	96.34	95.59	76.03	99.92	99.50	**100.00**	96.28	99.34	94.79	100.00	100.00	100.00	95.90
6	65.18	93.76	98.59	95.92	96.68	97.29	80.05	100.00	88.74	100.00	**100.00**	72.59	96.03	98.84	97.55	95.71	99.12	95.98	100.00	97.36	100.00	**100.00**
7	94.27	93.86	100.00	94.59	100.	97.98	99.73	100.00	97.46	100.00	**100.00**	96.62	93.23	100.00	99.85	99.54	100.00	100.00	100.00	100.00	100.00	**100.00**
8	89.25	98.77	**100.00**	98.88	99.38	97.66	95.46	98.68	87.98	99.13	99.45	89.05	98.39	99.85	98.11	99.82	99.96	96.7	100.00	93.78	**100.00**	99.90
9	96.27	95.59	100.00	99.64	94.73	99.66	87.06	**100.00**	99.85	100.00	98.96	93.73	97.85	100.00	99.94	100.00	**100.00**	90.38	98.63	100.00	98.83	99.96
10	89.2	99.17	100.00	99.56	99.71	100.	98.16	100.00	95.34	93.77	**100.00**	93.05	99.95	99.89	99.36	97.47	100.00	99.92	100.00	99.72	100.00	**100.00**
11	97.53	95.62	99.51	99.97	98.89	**100.00**	96.07	100.00	94.44	100.00	99.82	99.08	94.49	99.34	100.00	99.53	100.00	97.55	100.00	98.68	99.60	**100.00**
12	86.55	99.00	99.98	98.68	96.19	98.36	90.36	100.00	97.06	95.88	**100.00**	86.11	100.00	99.89	99.96	97.11	99.4.00	90.73	100.00	88.24	99.78	**100.00**
13	97.99	100.00	100.00	100.00	97.15	100.00	86.32	98.84	100.00	100.00	**100.00**	97.97	100.00	100.00	100.00	99.52	100.00	93.62	99.82	100.00	100.00	**100.00**
OA	89.43	95.85	99.44	99.24	97.02	99.25	91.14	99.42	94.04	98.89	**99.70**	90.01	97.80	99.52	99.66	97.80	99.77	94.93	99.77	95.94	99.62	**99.85**
AA	85.84	94.45	99.16	98.83	96.82	99.03	91.23	99.4	94.18	98.83	**99.52**	87.86	96.99	99.37	99.58	97.62	**99.68**	95.30	99.84	96.25	99.54	99.63
Kappa	88.2	95.36	99.37	99.15	96.67	99.16	90.12	99.35	93.35	98.76	**99.66**	88.85	97.53	99.47	99.62	97.53	99.79	94.32	99.74	95.46	99.57	**99.83**

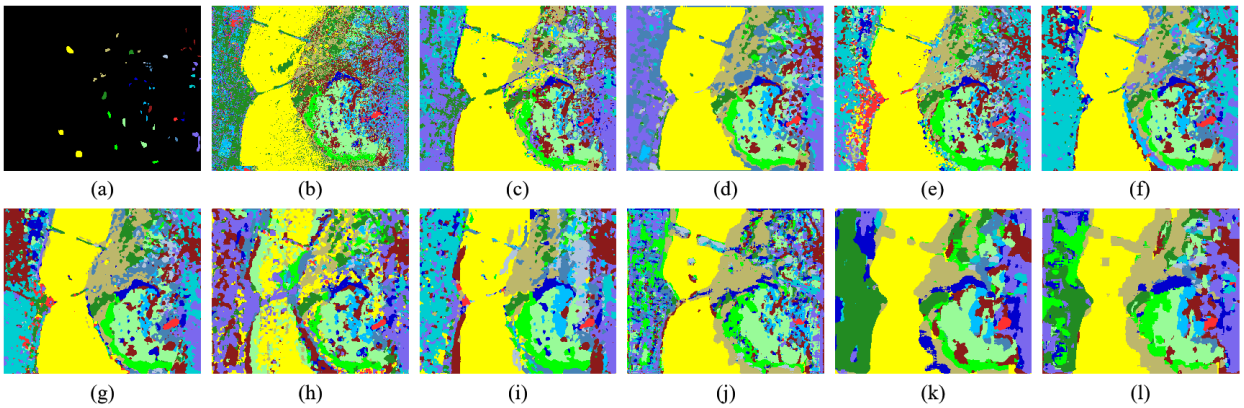


Fig. 8. Comparison of experimental results on KSC using 30 training samples in each category. (a) Ground-truth map. (b) 1D-AT-CNN, window size (WS) = 1. (c) 3D-ANAS, WS = 1. (d) LMAFNet, WS = 9. (e) A^2S^2K-RSNet, WS = 9. (f) EESINet, WS = 9. (g) LDWNet, WS = 9. (h) LS^{2}CM, WS = 11. (i) S3Net, WS = {13,7}. (j) 3D-LWNet, WS = 27. (k) 3D-AT-CNN, WS = 32. (l) LMSS-NAS, WS = 27.

TABLE VII

EXPERIMENTAL RESULTS ON THE PU DATASET

Models	1D-AT -CNN	3D-ANAS	LMAF -Net	A^2S^2K -RSNet	EESI -Net	LDW -Net	LS^{2}CM	S3Net	3D-LW -Net	3D-AT -CNN	LMSS -NAS	1D-AT -CNN	3D-ANAS	LMAF -Net	A^2S^2K -RSNet	EESI -Net	LDW -Net	LS^{2}CM	S3Net	3D-LW -Net	3D-AT -CNN	LMSS -NAS
Set.	20 labeled samples/class											30 labeled samples/class										
1	69.69	93.21	95.82	87.7	91.58	96.29	53.06	97.54	82.43	88.24	**97.60**	76.43	93.21	97.38	93.91	93.07	98.68	63.16	95.29	82.32	89.7	**99.58**
2	76.37	99.10	94.25	96.80	80.75	98.19	70.36	92.23	84.76	90.72	**99.54**	81.90	98.50	95.96	98.07	83.50	**99.58**	77.20	94.34	88.84	97.92	99.29
3	73.43	**99.28**	90.06	72.73	79.20	86.69	80.82	82.98	76.88	92.11	94.61	74.04	**99.61**	94.57	76.71	87.18	94.37	81.20	90.70	83.74	92.31	98.13
4	90.2	**99.22**	94.85	89.04	95.80	97.30	78.6	97.04	91.45	81.27	95.28	93.38	98.09	96.65	92.36	95.29	**98.52**	79.68	97.11	94.11	71.22	97.32
5	96.54	**100.00**	99.19	97.03	99.33	99.64	98.16	99.85	96.23	93.12	98.64	96.13	**100.00**	99.75	96.09	99.46	99.83	99.10	99.77	96.90	95.12	99.05
6	75.48	97.36	93.40	76.11	91.74	88.92	73.50	93.48	92.50	**98.47**	89.66	81.81	**99.80**	98.98	86.27	87.20	95.48	69.94	90.78	92.10	96.96	97.36
7	88.83	99.62	**99.65**	78.13	85.52	94.89	92.79	98.74	93.56	96.14	89.09	88.23	**99.77**	99.82	80.09	89.36	95.46	94.62	99.69	95.46	95.99	95.50
8	77.59	87.35	94.47	64.70	78.06	78.52	40.54	92.93	96.01	96.84	**97.99**	74.05	96.55	94.16	73.13	74.08	90.54	46.22	92.04	93.19	94.98	**96.58**
9	96.66	98.69	**99.66**	83.94	99.30	96.72	82.44	95.7	89.16	79.62	95.74	95.65	**100.00**	99.05	93.72	98.85	98.13	85.64	97.03	92.43	80.64	98.53
OA	77.59	96.53	94.69	85.29	85.62	94.04	68.38	93.67	87.10	91.16	**96.78**	81.75	97.81	96.66	90.02	86.63	97.63	73.26	94.28	89.25	93.36	**98.52**
AA	82.75	**96.31**	95.71	82.91	89.03	93.02	74.48	94.50	89.22	90.72	95.35	84.62	**98.39**	97.37	87.82	89.78	96.69	77.42	95.19	91.01	90.54	97.93
Kappa	71.51	95.39	93.01	80.9	81.61	92.13	60.46	91.68	83.35	88.51	**95.75**	76.55	97.1	95.6	87.07	82.99	96.86	65.83	92.46	86.05	91.50	**98.03**

using spectral–spatial features, 1D-AT-CNN is the worst as it only utilized spectral features. As shown in Table VIII, when training samples per class are increased to 30 on PC, compared with the NAS-based model 3D-ANAS (99.47%), the proposed method is suboptimal (99.38%). Meanwhile, for other methods, LMSS-NAS improves the classification results in terms of OA and Kappa by at least 0.05% and 0.07%, respectively. Also, the highest and the lowest of LMSS-NAS over ten replicates are up to 99.50% and 99.28%, respectively.

From the classification results of the four benchmark datasets, it can be seen that the overall performance of the NAS-based methods is better than those of the handcrafted methods, which fully demonstrates the adaptability of NAS to different datasets. Among the NAS-based methods, the 3D-AT-CNN, 3D-ANAS, and the proposed LMSS-NAS, which consider both spectral and spatial information simultaneously, are significantly better than 1D-AT-CNN, which considers the spectral and spatial information separately. Furthermore, from the model based on the spectral–spatial information, it can be observed that the lightweight methods with the attention mechanism, such as LMSS-NAS, LMAFNet, and A^2S^2K-RSNet, have higher overall classification accuracy than the lightweight models 3D-LWNet, LS^{2}CM, S3Net, EESINet, and LDWNet without attention. Also, our model LMSS-NAS is higher than LMAFNet and A^2S^2K-RSNet, indicating that our method can better model the spectral–spatial features by the M_{CS} attention. To further demonstrate the robustness of the proposed method, the standard deviations are presented in Table IX.

Meanwhile, to verify the generalization of LMSS-NAS, we use the cells architecture searched from HU to conduct classification training on KSC, PU, and PC. As shown in Table X, LMSS-NAS also achieves good performance and even outperforms handcrafted models on KSC and PU, which demonstrates the searched architecture's generalization.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:27:38 UTC from IEEE Xplore. Restrictions apply.

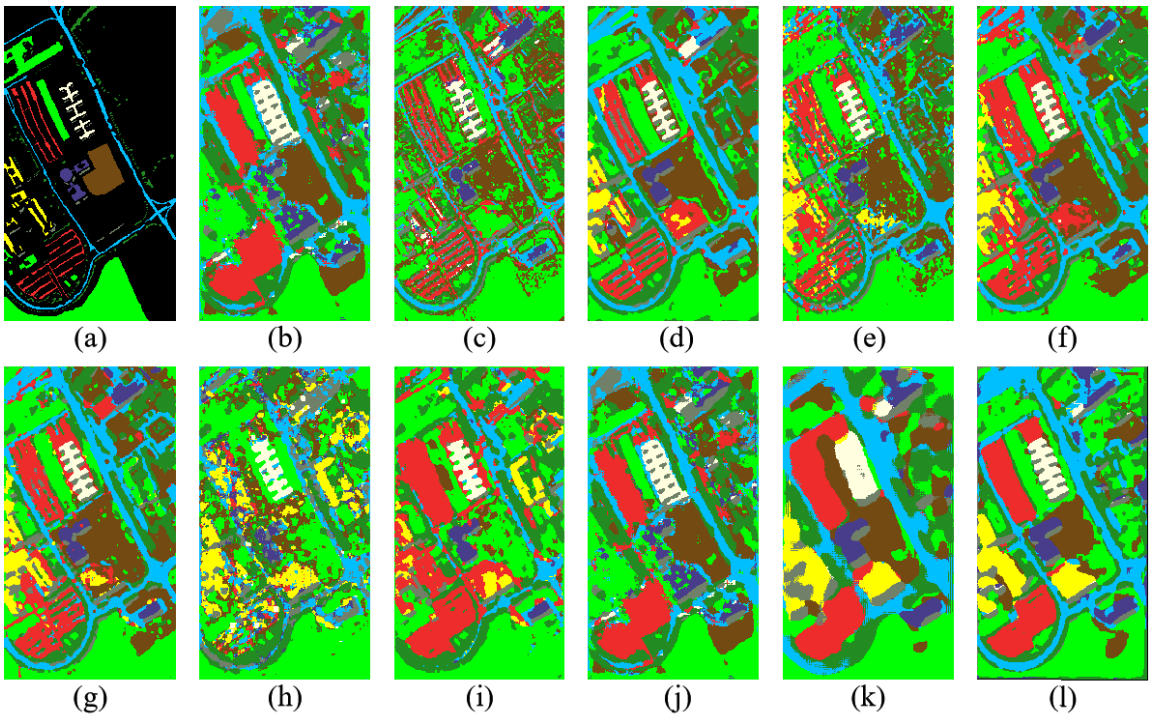


Fig. 9. Comparison of experimental results on PU using 30 training samples in each category. (a) Ground-truth map. (b) 1D-AT-CNN, window size (WS) = 1. (c) 3D-ANAS, WS = 1. (d) LMAFNet, WS = 9. (e) A^2S^2K-RSNet, WS = 9. (f) EESINet, WS = 9. (g) LDWNet, WS = 9. (h) LS^2CM, WS = 11. (i) S3Net, WS = {13,7}. (j) 3D-LWNet, WS = 27. (k) 3D-AT-CNN, WS = 32. (l) LMSS-NAS, WS = 27.

D. Model Complexity

As this article aims to design a lightweight HSI automatic classification model, we compare the complexity with state of the art in terms of required number of model parameters (Params), FLOPs, and GPU memory usage (GPU).

As shown in Table XI, it can be seen that the different search space operations and different network architecture jointly result in the difference of the parameter amount of the networks. The FLOPs are determined by both the input data size and the parameters. In the network training, the GPU memory usage is associated with the batch size, parameters, input data size, and the model's architectural parameters. Therefore, even if parameters are small, the larger batch size will result in larger GPU memory usage. For example, the parameters and FLOPs of the 1D-AT-CNN, LS^2CM, EESINet, and LDWNet models are extremely small, but the GPU memory usage is relatively larger and the classification accuracy is lower.

Therefore, it is necessary to combine the network model complexity and classification performance to compare and analyze. From the result of four datasets, as we can observe, compared with other methods, the proposed requires the lighter number of Params, FLOPs, and GPU memory usage, while achieving the second best (on PC) or even the best (on HU, KSC, and PU) classification results. Benefiting from the effectiveness of our unique lightweight search space design and the adaptability of the search architecture to different datasets, for handcrafted lightweight models, whether pure lightweight convolutional designed models 3D-LWNet, LS^2CM, S3Net, EESINet, and LDWNet, or the models of LMAFNet and A^2S^2K-RSNet with added attention mechanism, the proposed LMSS-NAS not only achieves the highest OA but also lightweight Params, FLOPs, and GPU usage. Meanwhile, for NAS-based searched lightweight models 1D-AT-CNN, 3D-AT-CNN, and 3D-ANAS, compared with the models of 1D-AT-CNN and 3D-AT-CNN, which used 1-D convolution and 2-D convolution, respectively, the proposed LMSS-NAS has similar occupancy in terms of Params, FLOP, and GPU, but gets the highest OA. Compared to 3D-ANAS with 3-D convolution, on HU, KSC, and PU, LMSS-NAS achieves the best OA while occupying half or even less number of Params, FLOPs, and GPU. We also compared the model complexity under the same window size with the competitors (the details can be seen in Tables XI–XVII in the Supplementary Material).

At the same time, we compare the test time and inference speed of each model on the four datasets. The test computational time of models on four datasets is summarized in Fig. 11. Combining with the overall results, it can be seen that NAS-based methods generally cost less test time compared to handcrafted methods. The reason is that NAS-based methods design a more suitable lightweight model for HSI classification. Meanwhile, the test time varies with the total test samples that are different under different sampling methods. Accordingly, compared to the models that use the overlap sampling, such as 3D-LWNet, 3D-AT-CNN, LMAFNet, A^2S^2K-RSNet, LS^2CM, S3Net, EESINet, and LDWNet, the proposed LMSS-NAS achieves the best classification performance with less test time, while it requires a slightly higher test time than the models without using overlap sampling, such as 1D-AT-CNN and 3D-ANAS.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:27:38 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII

EXPERIMENTAL RESULTS ON THE PC DATASET

Models	1D-AT -CNN	3D-ANAS	LMAF -Net	A^2S^2K -RSNet	EESI -Net	LDW -Net	LS^{2}CM	S3Net	3D-LW -Net	3D-AT -CNN	LMSS -NAS	1D-AT -CNN	3D-ANAS	LMAF -Net	A^2S^2K -RSNet	EESI -Net	LDW -Net	LS^{2}CM	S3Net	3D-LW -Net	3D-AT -CNN	LMSS -NAS
Set.	20 labeled samples/class											30 labeled samples/class										
1	99.81	**100.00**	99.21	99.92	99.22	99.94	94.32	99.71	99.61	99.56	99.92	99.76	99.94	99.24	99.99	99.47	**100.00**	95.61	99.76	99.52	99.78	99.98
2	80.90	94.66	96.77	97.46	91.94	**97.90**	81.23	96.31	91.85	87.79	95.66	87.90	95.49	97.11	97.71	92.18	98.45	77.73	96.29	93.92	92.48	**98.83**
3	89.04	96.21	**97.67**	84.57	96.35	81.29	80.55	95.69	86.77	82.98	94.02	91.16	96.73	97.98	86.95	96.58	88.66	87.28	**97.63**	89.26	85.81	95.29
4	73.53	**99.89**	99.79	88.00	97.22	89.56	77.28	99.73	92.85	98.85	96.78	79.72	99.62	**99.98**	95.33	94.54	98.34	90.14	99.37	91.10	97.32	98.43
5	90.05	97.60	84.97	94.69	92.10	**98.61**	73.74	95.72	95.53	95.83	94.73	92.10	99.69	98.93	97.38	93.29	**99.78**	77.45	98.43	96.09	97.02	97.82
6	95.53	**99.80**	98.95	90.76	91.17	93.35	52.35	89.65	80.97	93.22	97.39	96.47	**99.35**	98.79	94.73	97.09	96.86	60.83	89.28	90.73	95.91	97.30
7	84.23	97.88	93.21	99.37	90.50	**99.89**	87.64	96.68	85.66	94.94	98.13	85.43	97.55	96.48	99.93	90.85	99.95	84.96	97.81	93.24	95.29	**99.80**
8	98.17	99.25	98.07	**99.94**	96.84	99.88	64.44	98.72	85.06	95.12	99.39	97.88	99.90	99.38	**99.95**	97.27	99.91	67.18	99.39	87.32	95.13	99.82
9	98.89	**99.96**	99.48	88.93	95.80	77.93	81.69	97.27	91.34	85.29	93.81	98.58	**100.00**	99.22	95.87	95.14	96.56	85.91	97.79	93.7	91.84	95.39
OA	96.18	**99.21**	97.8	98.06	96.75	98.05	80.31	98.17	92.42	96.25	98.78	96.79	**99.47**	98.98	98.93	97.38	99.33	82.52	98.58	94.22	96.99	99.38
AA	90.02	**98.36**	96.46	93.74	94.57	93.15	77.03	96.61	89.96	92.62	96.65	92.11	**98.70**	98.57	96.43	95.16	97.61	80.79	97.30	92.76	94.51	98.07
Kappa	94.6	**98.88**	96.89	97.25	95.42	97.24	73.31	97.4	89.41	94.71	98.27	95.47	**99.24**	98.56	98.48	96.31	99.05	76.08	97.98	91.92	95.76	99.12

TABLE IX

OVERALL STANDARD DEVIATIONS OF LMSS-NAS IN FOUR HSI DATASETS

Dataset	HU	KSC	PU	PC	HU	KSC	PU	PC
Set.	20 labeled samples/class				30 labeled samples/class			
OA	91.84±0.22	99.70±0.26	96.78±0.47	98.78±0.17	94.61±0.25	99.85±0.07	98.52±0.35	99.38±0.08
AA	92.06±0.22	99.52±0.19	95.35±0.76	96.65±0.37	94.27±0.26	99.63±0.18	97.93±0.35	98.07±0.19
Kappa	91.18±0.24	99.66±0.29	95.75±0.61	98.27±0.24	94.17±0.27	99.83±0.07	98.03±0.47	99.12±0.11

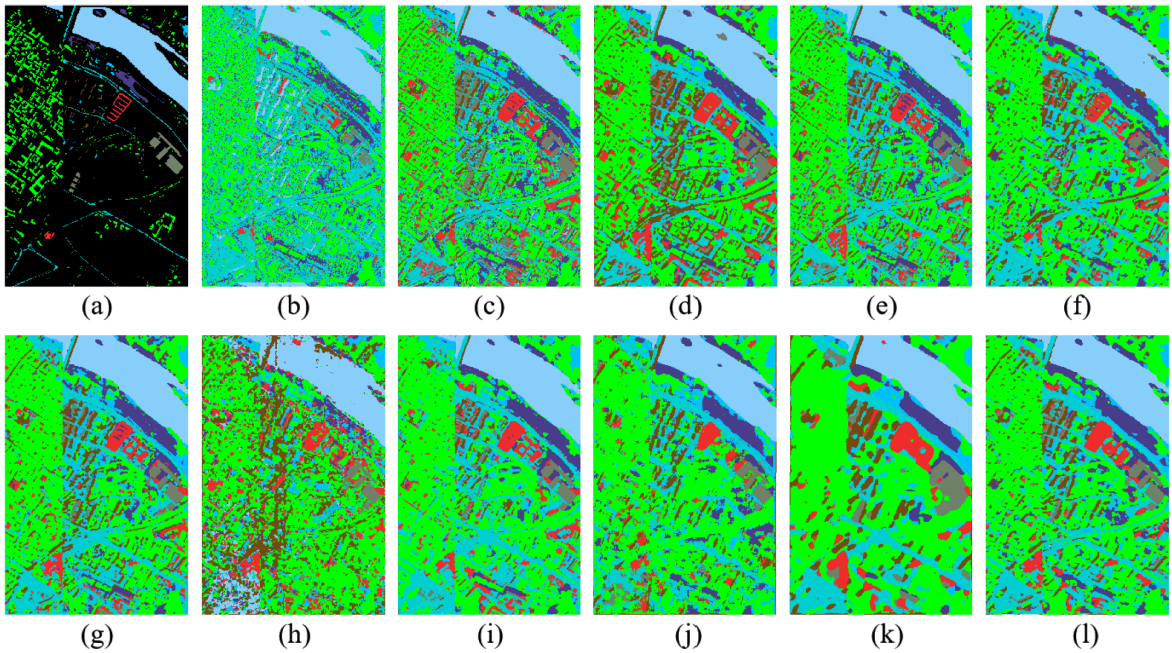


Fig. 10. Comparison of experimental results on PC using 30 training samples in each category. (a) Ground-truth map. (b) 1D-AT-CNN, window size (WS) = 1. (c) 3D-ANAS, WS = 1. (d) LMAFNet, WS = 9. (e) A^2S^2K-RSNet, WS = 9. (f) EESINet, WS = 9. (g) LDWNet, WS = 9. (h) LS^{2}CM, WS = 11. (i) S3Net, WS = {13,7}. (j) 3D-LWNet, WS = 27. (k) 3D-AT-CNN, WS = 32. (l) LMSS-NAS, WS = 27.

TABLE X

MODEL GENERALIZATION EXPERIMENT

Data set	KSC	PU	PC	KSC	PU	PC
Set.	20 labeled samples/class			30 labeled samples/class		
OA	99.34±0.10	92.83±0.38	97.60 ±0.13	99.68±0.08	97.18±0.34	98.20±0.17
AA	99.18±0.15	87.36±0.66	92.59±0.41	99.28±0.17	94.62±0.60	94.45±0.50
Kappa	99.26±0.11	90.55±0.49	96.60±0.18	99.65±0.08	96.26±0.45	97.46±0.23

The inference speed (KP/S) is obtained by $N \times N \times \mathrm{TP}/T/1000$, where T (S) is the time required for the model to process the test samples, $N \times N$ is the spatial patch size repeatedly calculated for each labeled sample, and TP is the number of labeled samples to be tested. Here, $N = 1$ if the overlap sampling mode is not used. Fig. 12 shows the superiority of the proposed method in terms of inference speed. As shown in Fig. 12, the inference speed of the proposed method is higher than most of competitors on the four datasets except for the 3D-AT-CNN on the KSC dataset. However, it is worth noting that ours is comparable as the proposed method achieves the highest OA, the relatively lower Params, FLOPs, and GPU memory usage on KSC.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:27:38 UTC from IEEE Xplore. Restrictions apply.

TABLE XI
COMPARISON OF COMPUTING RESOURCES OF DIFFERENT ALGORITHMS ON FOUR DATASETS

Data set	HU				KSC			
Models	Params/M	FLOPs/G	GPU/G	OA/%	Params/M	FLOPs/G	GPU/G	OA/%
1D-AT-CNN [44]	0.0687	0.0009	0.599	58.35	0.0524	0.0007	0.593	89.43
3D-ANAS [45]	0.2267	3.0934	3.755	83.15	0.2243	3.7915	5.071	95.85
LMAFNet [36]	0.1563	0.0003	3.018	90.67	0.1581	0.0003	3.018	99.44
A^2S^2K-RSNet [34]	0.2833	0.1219	0.843	90.94	0.3332	0.1497	0.929	99.24
EESINet [27]	0.0799	0.0066	1.455	85.37	0.1096	0.009	1.265	97.02
LDWNet [28]	0.0396	0.0347	1.527	91.31	0.0463	0.0426	1.769	99.25
LS^2CM [26]	0.0116	0.0013	0.835	80.31	0.0120	0.0019	0.763	91.14
S3Net [25]	0.3267	0.1741	4.997	98.17	0.3917	0.2115	3.615	99.42
3D-LWNet [24]	1.0407	0.7673	6.351	78.95	1.0401	0.9472	7.737	94.04
3D-AT-CNN [44]	0.1091	0.0290	1.119	83.37	0.0689	0.0265	1.135	98.89
LMSS-NAS	0.0568	0.0147	1.009	91.84	0.0854	0.0187	1.095	99.70
Data set	PU				PC			
Models	Params/M	FLOPs/G	GPU/G	OA/%	Params/M	FLOPs/G	GPU/G	OA/%
1D-AT-CNN	0.0725	0.0009	0.599	77.59	0.0490	0.0007	0.595	96.18
3D-ANAS	0.2156	2.0088	3.711	96.53	0.2160	1.9804	3.177	99.21
LMAFNet	0.1529	0.0003	3.018	94.69	0.1529	0.0003	3.018	97.80
A^2S^2K-RSNet	0.2207	0.0873	0.767	85.29	0.2177	0.0856	0.767	98.06
EESINet	0.0477	0.004	1.281	85.62	0.0473	0.0039	1.279	96.75
LDWNet	0.0305	0.0247	1.263	94.04	0.0301	0.0242	1.243	98.05
LS^2CM	0.0106	0.0017	0.823	68.38	0.0106	0.0017	0.821	80.31
S3Net	0.0915	0.0399	1.431	93.67	0.0915	0.0399	1.431	98.17
3D-LWNet	1.0391	0.5400	4.683	87.10	1.0391	0.5340	4.639	92.42
3D-AT-CNN	0.0537	0.0198	1.003	91.16	0.0979	0.0261	1.123	96.25
LMSS-NAS	0.1051	0.0192	1.167	96.78	0.0968	0.0171	1.053	98.78

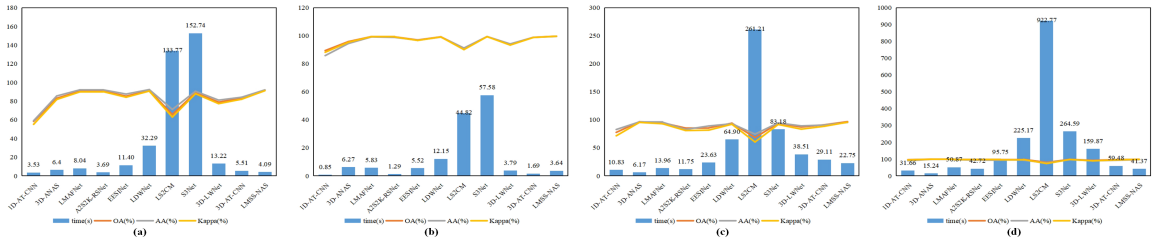


Fig. 11. Comparison of test time of each model in different datasets. (a) HU. (b) KSC. (c) PU. (d) PC.

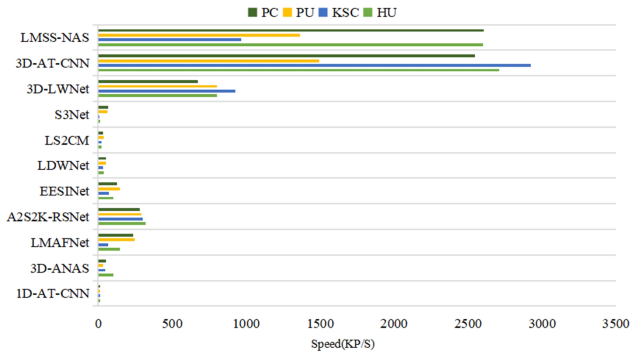


Fig. 12. Comparison of inference speed of each model in different datasets.

E. Ablation Study

1) Effectiveness of the Search Space: In this article, the cells searched by LMSS-NAS are composed of a combination of candidate operations in the search space. By searching for the optimal operation combination in the cell, we can effectively learn data features adaptively and improve the classification accuracy. To verify whether LMSS-NAS improves the accuracy by searching for efficient architectures or simply by integrating various branching and convolution operations, we manually modify the architecture searched by LMSS-NAS in three different ways.

Taking the architecture on HU dataset as an example, as shown in Figs. 13 and 14, the modified operations in normal and reduction cells are marked in red. Specifically, the three modifications include: 1) randomly replacing two selected operations with others in the cell (termed LMSS-NAS-Rop); 2) changing the connection relationships between the nodes inside each cell (termed LMSS-NAS-Rtopo); and 3) mixing the designed operations with the depth SC to expand the search space (termed LMSS-NAS-Rbop). From Table XII, it can be seen that the performance of the manually modified structure is lower than that of the automatically searched architecture, which shows the excellent performance of NAS for the automatic design of model architecture. Meanwhile, when increasing the depth separate convolution into the search

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:27:38 UTC from IEEE Xplore. Restrictions apply.

TABLE XII
EFFECTIVENESS OF LMSS-NAS

Models	LMSS-NAS	LMSS-NAS-Rop	LMSS-NAS- Rtopo	LMSS-NAS-Rbop
OA	94.61±0.25	93.30±0.56	93.33±0.36	94.16±0.43
AA	94.27±0.26	93.08±0.69	93.21±0.33	93.78±0.41
Kappa	94.17±0.27	92.76±0.60	92.79±0.39	93.69±0.47

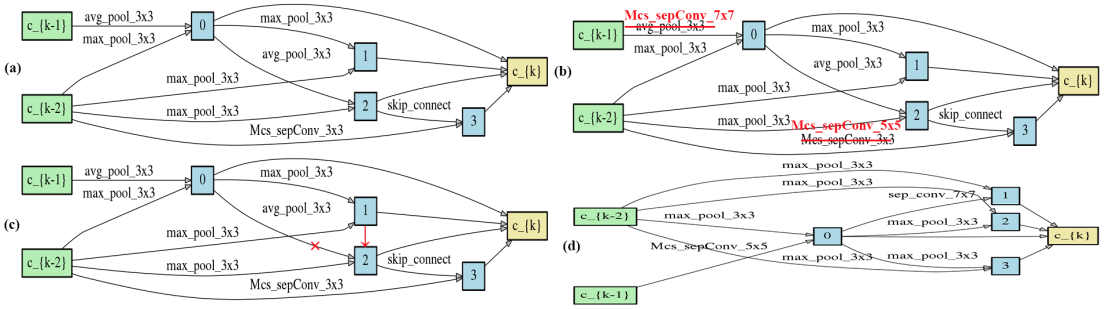


Fig. 13. Normal cell. (a) LMSS-NAS. (b) LMSS-NAS-Rop. (c) LMSS-NAS-Rtopo. (d) LMSS-NAS-Rbop.

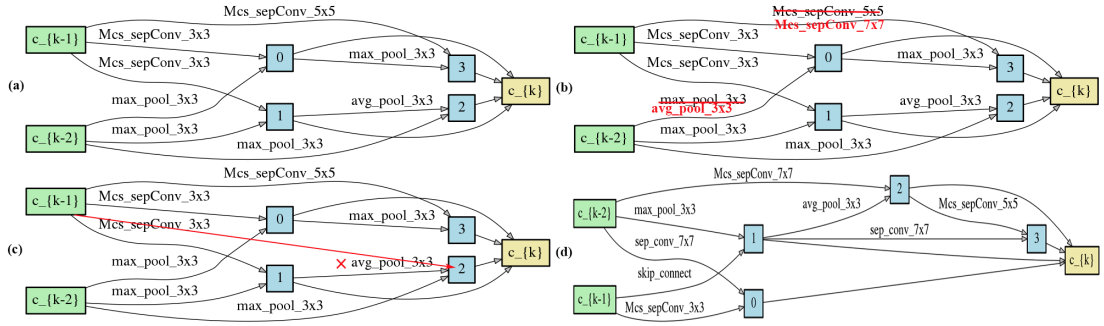


Fig. 14. Reduction cell. (a) LMSS-NAS. (b) LMSS-NAS-Rop. (c) LMSS-NAS-Rtopo. (d) LMSS-NAS-Rbop.

space, the searched cell shows that the designed operator is more inclined to be chosen than other convolution operators.

2) Robustness and Model Regularization: They are given as follows.

a) Robustness: As the experiment in this article was based on a small sample classification, in order to verify the stability of the model, we increase the number of training samples to observe the model performance. All experiments were carried out ten times to obtain the average value. As can be seen from Fig. 15, our proposed method shows significant advantages.

b) Model regularization: To verify the spectral–spatial attention M_{CS} and fusion loss introduced in this article, we present the ablation validations, as shown in Table XIII. Here, the validation experiments include: 1) the convolution operator M_{CS}_sepConv_$(k \times k)(k = 3, 5, 7)$ without attention M_{CS} and CE loss; 2) the convolution operator M_{CS}_sepConv_$(k \times k)$ and CE loss (CE + M_{CS}); 3) the convolution operator M_{CS}_sepConv_$(k \times k)$ and FL (FL + M_{CS}); 4) the convolution operator M_{CS}_sepConv_$(k \times k)$ and label smoothing loss (SM + M_{CS}); 5) the convolution operator M_{CS}_sepConv_$(k \times k)$ without attention M_{CS} and fusion loss (FL + SM); and 6) the convolution operator M_{CS}_sepConv_$(k \times k)$ and fusion loss, which is the proposed model LMSS-NAS. It can be seen from Table XIII that CE achieves better classification performance compared with the lightweight models 3D-LWNet, LS^2CM, and Auto-CNN with the same loss function, which shows that the proposed equivalent stacked SSC operations are better in classification performance and lightweight. Also, compared with the models A^2S^2K-RSNet, S3Net, and LMAFNet that also combine convolution with attention mechanism, CE + M_{CS} has better

TABLE XIII
COMPARATIVE EXPERIMENTS OF FUSION LOSS AND SPATIAL–SPECTRAL ATTENTION

Models	CE	CE+M_{cs}	FL+M_{cs}	SM+M_{cs}	FL+SM	LMSS-NAS
1	94.90	97.00	96.75	97.66	96.86	97.72
2	88.82	84.41	89.54	96.82	89.55	96.28
3	99.00	99.39	94.88	99.18	99.31	99.30
4	80.49	92.29	83.81	95.74	80.29	91.02
5	97.25	96.66	98.77	97.01	96.39	97.32
6	82.72	81.83	82.34	75.42	79.51	81.91
7	85.30	89.64	91.68	92.65	86.81	92.29
8	93.31	95.41	98.89	98.50	99.93	98.56
9	84.89	79.93	86.92	90.02	85.54	88.76
10	83.36	81.42	83.60	86.91	86.90	91.98
11	97.46	97.36	98.29	98.30	98.59	97.18
12	96.19	95.40	94.80	98.29	98.12	97.59
13	96.14	93.57	95.48	90.45	91.88	91.77
14	94.04	99.25	95.41	96.82	100.00	97.59
15	93.37	95.83	95.28	95.63	90.14	94.61
OA	90.70	91.14	92.06	94.61	91.68	94.6
AA	91.15	91.96	92.43	93.96	91.99	94.26
Kappa	89.95	90.42	91.41	94.18	91.01	94.16

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:27:38 UTC from IEEE Xplore. Restrictions apply.

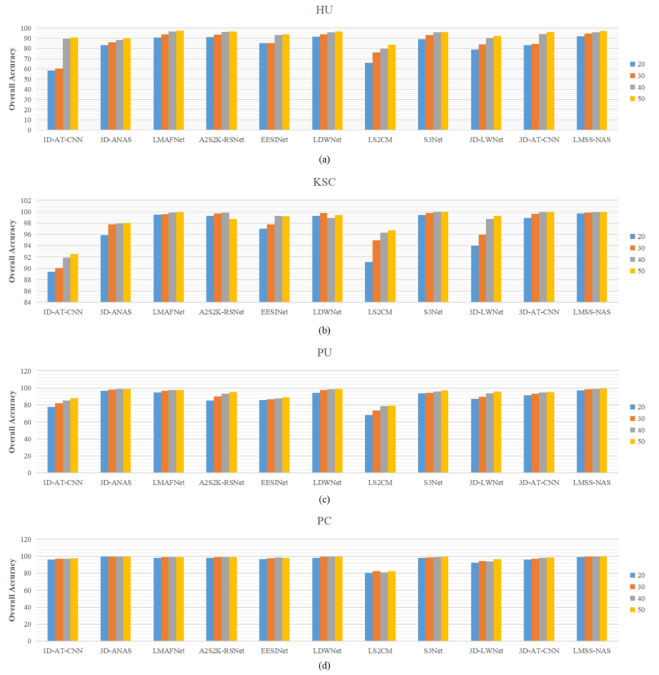


Fig. 15. Classification accuracy histograms of different training samples on four datasets. (a) HU. (b) KSC. (c) PU. (d) PC.

classification performance, which verifies the superiority of the attention M_{CS}.

IV. Conclusion

This article proposes a lightweight model for HSI classification where an SSC operator combined with an attention mechanism is designed to construct the search space, and a differentiable NAS method is used to search the network automatically. Experimental results show that the lightweight search space design reduces the number of network parameters and avoids the tedious process of manual model design and parameter tuning. By combining FL and label smoothing loss, the classification imbalance between classes is alleviated in a certain extent and the robustness of the model is enhanced as well. In addition, the architecture is optimized without manual design, and LMSS-NAS achieves a better classification performance with a lower computational cost. It shows that the lightweight NAS method has great potential for accurate and fast HSI classification.

Acknowledgment

The authors would like to thank Dr. H. Zhang for sharing the code of 3-D asymmetric neural architecture search (3D-ANAS) and 3-D lightweight convolution network (3D-LWNet) algorithms, Prof. Y. Chen for 1D-AT-CNN and 3D-AT-CNN, Prof. S.K.Roy for A^2S^2K-RSNet, Dr. J. Wang for LMAFNet, Dr. Z. Xue for S3Net, Dr. Z. Meng for LS^2CM, and PD. X. -M. Dong for EESINet and LDWNet.

References

[1] B. Zhang, "Advancement of hyperspectral image processing and information extraction," *J. Remote Sens.*, to be published.

[2] X. Yang and Y. Yu, "Estimating soil salinity under various moisture conditions: An experimental study," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2525–2533, May 2017.

[3] W. Zhao and S. Du, "Spectral–spatial feature extraction for hyperspectral image classification: A dimension reduction and deep learning approach," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4544–4554, Apr. 2016.

[4] M. S. Alias, W. A. Wan Ghazali, and A. Puteri Arlydia, "Aerial inventory of fragmented forest using airborne hyperspectral imaging," in *Proc. 4th Workshop Hyperspectral Image Signal Process. (WHISPERS)*, Jun. 2012, pp. 1–5.

[5] C. Li, S. Li, X. Kang, and T. Lu, "Gabor filtering based deep network for hyperspectral image classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 1808–1811.

[6] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.

[7] Y. Chen, X. Zhao, and X. Jia, "Spectral–spatial classification of hyperspectral data based on deep belief network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, pp. 2381–2392, 2015.

[8] W. Hu, Y. Huang, L. Wei, F. Zhang, and H. Li, "Deep convolutional neural networks for hyperspectral image classification," *J. Sensors*, vol. 2015, Jul. 2015, Art. no. 258619.

[9] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.

[10] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.

[11] A. Sellami, M. Farah, I. R. Farah, and B. Solaiman, "Hyperspectral imagery classification based on semi-supervised 3-D deep neural network and adaptive band selection," *Expert Syst. Appl.*, vol. 129, pp. 246–259, Sep. 2019.

[12] X. Kang, B. Zhuo, and P. Duan, "Dual-path network-based hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 3, pp. 447–451, Mar. 2019.

[13] J. Li, X. Zhao, Y. Li, Q. Du, B. Xi, and J. Hu, "Classification of hyperspectral imagery using a new fully convolutional neural network," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 2, pp. 292–296, Feb. 2018.

[14] A. B. Hamida, A. Benoit, P. Lambert, and C. B. Amar, "3-D deep learning approach for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4420–4434, Aug. 2018.

[15] J. W. P. Sun, H. Li, W. Li, X. Meng, C. Ge, and Q. Du, "Low-rank and sparse representation for hyperspectral image processing: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 1, pp. 10–43, Jun. 2021.

[16] M. He, B. Li, and H. Chen, "Multi-scale 3D deep convolutional neural network for hyperspectral image classification," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2017, pp. 3904–3908.

[17] H. Lee and H. Kwon, "Going deeper with contextual CNN for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 26, no. 10, pp. 4843–4855, Oct. 2017.

[18] Q. Hao, S. Li, L. Fang, and X. Kang, "Multiscale feature extraction with Gaussian curvature filter for hyperspectral image classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Sep. 2020, pp. 80–83.

[19] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D-2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Jun. 2020.

[20] C. Szegedy et al., "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 1–9.

[21] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the inception architecture for computer vision," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2818–2826.

[22] K. Han, Y. Wang, Q. Tian, J. Guo, C. Xu, and C. Xu, "GhostNet: More features from cheap operations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 1577–1586.

[23] S. Jia et al., "A lightweight convolutional neural network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4150–4163, May 2021.

[24] H. Zhang, Y. Li, Y. Jiang, P. Wang, Q. Shen, and C. Shen, "Hyperspectral classification based on lightweight 3-D-CNN with transfer learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5813–5828, Aug. 2019.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:27:38 UTC from IEEE Xplore. Restrictions apply.

[25] Z. Xue, Y. Zhou, and P. Du, "S3Net: Spectral–spatial Siamese network for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5531219.
[26] Z. Meng, L. Jiao, M. Liang, and F. Zhao, "A lightweight spectral–spatial convolution module for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[27] Z. Lv, X.-M. Dong, J. Peng, and W. Sun, "ESSINet: Efficient spatial–spectral interaction network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5525715.
[28] B. Cui, X.-M. Dong, Q. Zhan, J. Peng, and W. Sun, "LiteDepthwiseNet: A lightweight network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.
[29] L. Chen et al., "SCA-CNN: Spatial and channel-wise attention in convolutional networks for image captioning," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 6298–6306.
[30] J. Hu, L. Shen, S. Albanie, G. Sun, and E. Wu, "Squeeze-and-excitation networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 8, pp. 2011–2023, Aug. 2020.
[31] S. Woo, J. Park, J. Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 3–19.
[32] M. Zhu, L. Jiao, F. Liu, S. Yang, and J. Wang, "Residual spectral–spatial attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 449–462, Jan. 2021.
[33] Z. Xue, M. Zhang, Y. Liu, and P. Du, "Attention-based second-order pooling network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9600–9615, Nov. 2021.
[34] S. K. Roy, S. Manna, T. Song, and L. Bruzzone, "Attention-based adaptive spectral–spatial kernel ResNet for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7831–7843, Sep. 2021.
[35] Z. Xue, Q. Xu, and M. Zhang, "Local transformer with spatial partition restore for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 4307–4325, 2022.
[36] J. Wang et al., "NAS-guided lightweight multiscale attention fusion network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8754–8767, Oct. 2021.
[37] I. Radosavovic, R. P. Kosaraju, R. Girshick, K. He, and P. Dollár, "Designing network design spaces," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10425–10433.
[38] Q. Wang, B. Wu, P. Zhu, P. Li, W. Zuo, and Q. Hu, "ECA-Net: Efficient channel attention for deep convolutional neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 11531–11539.
[39] C. Liu et al., "Auto-DeepLab: Hierarchical neural architecture search for semantic image segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2019, pp. 82–92.
[40] H. Zhang, Y. Li, H. Chen, and C. Shen, "Memory-efficient hierarchical neural architecture search for image denoising," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 3654–3663.
[41] P. Liu, M. D. El Basha, Y. Li, Y. Xiao, P. C. Sanelli, and R. Fang, "Deep evolutionary networks with expedited genetic algorithms for medical image denoising," *Med. Image Anal.*, vol. 54, pp. 306–315, May 2019.
[42] X. Chu, B. Zhang, H. Ma, R. Xu, J. Li, and Q. Li, "Fast, accurate and lightweight super-resolution with neural architecture search," in *Proc. 25th Int. Conf. Pattern Recognit. (ICPR)*, Jan. 2021, pp. 59–64, doi: 10.1109/ICPR48806.2021.9413080.
[43] H. Liu, K. Simonyan, and Y. Yang, "Darts: Differentiable architecture search," in *Proc. ICLR*, 2018, pp. 1–13.
[44] Y. Chen, K. Zhu, L. Zhu, X. He, P. Ghamisi, and J. A. Benediktsson, "Automatic design of convolutional neural network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 7048–7066, Sep. 2019.
[45] H. Zhang, C. Gong, Y. Bai, Z. Bai, and Y. Li, "3-D-ANAS: 3-D asymmetric neural architecture search for fast hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5508519.
[46] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 2, pp. 318–327, Feb. 2020.

Chunhong Cao received the B.S. degree in applied mathematics and software from Central South University, Changsha, China, in 1999, and the M.S. and Ph.D. degrees from Xiangtan University, Xiangtan, China, in 2005 and 2017, respectively.

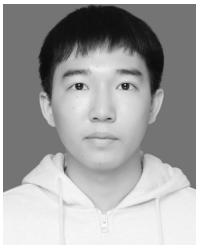
She was a Visiting Scholar with the Department of Computer Science, University of Georgia, Athens, GA, USA. She is currently a Professor with the School of Computer Science and the School of Cyberspace Security, Xiangtan University. Her recent research interests include hyperspectral image processing, machine learning, and compressive sensing.

Han Xiang received the B.E. degree from the Hunan Police College, Changsha, China, in 2020. She is currently pursuing the M.E. degree in computer technology with the School of Computer Science and the School of Cyberspace Security, Xiangtan University, Xiangtan, China.

Her research interests include hyperspectral image classification, machine learning, and deep learning.

Wei Song received the B.E. degree from the School of Computer Science and the School of Cyberspace Security, Xiangtan University, Xiangtan, China, in 2020. He is currently pursuing the M.E. degree in computer technology with the College of Information Engineering, Xiangtan University.

His research interests include hyperspectral image unmixing, pattern recognition, and machine learning.



Hongbo Yi (Member, IEEE) received the B.E. degree from the School of Computer Science and the School of Cyberspace Security, Xiangtan University, Xiangtan, China, in 2021, where he is currently pursuing the M.E. degree in computer technology.

Fen Xiao (Member, IEEE) received the B.S. and Ph.D. degrees from Xiangtan University, Xiangtan, China, in 2002 and 2008, respectively.

She was a Visiting Scholar at the State Key Laboratory of Pacific Northwest Pacific, Richland, WA, USA. She is currently a Professor with the School of Computer Science, School of Cyberspace Security, Xiangtan University. Her current research interests include visual saliency detection, remote sensing, image analysis, and image description.

Xieping Gao (Member, IEEE) received the B.S. and M.S. degrees from Xiangtan University, Xiangtan, China, in 1985 and 1988, respectively, and the Ph.D. degree from Hunan University, Changsha, China, in 2003.

He was a Visiting Scholar with the National Key Laboratory of Intelligent Technology and Systems, Tsinghua University, Beijing, China, from 1995 to 1996, and the School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore, from 2002 to 2003. He is currently a Professor with the Hunan Provincial Key Laboratory of Intelligent Computing and Language Information Processing, Hunan Normal University, Changsha. His research interests include the areas of neural networks, evolution computation, and hyperspectral image processing.

Dr. Gao is a regular reviewer for several journals and he has been a member of the technical committees of several scientific conferences.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:27:38 UTC from IEEE Xplore. Restrictions apply.